

# Supplement: Gaming ISO 20022 Payment Messages

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**Abstract**—This supplement carries the material the five-page paper refers to: generator distributions for configurations A and B, screening-abstraction constants and the calibration grid, feature and detector definitions with hyperparameters, the complete test-window table for all eight detectors, per-primitive counts and paired SEL, the full pre-specified contradiction test, the defect-rate sweep, ablations, prevalence sensitivity, the generator shift for all detectors, XSD validation by primitive, the artefact audit, the leave-one-family-out mechanism, the release statement, and the domain-rater questionnaire. Every number is filled from `results/summary.json` at commit `17bb868dd2b1`.

## I. GENERATOR

TABLE I  
GENERATOR CONFIGURATIONS. B IS USED ONLY AS A TEST DISTRIBUTION IN THE SHIFT EXPERIMENT.

| Parameter             | A (main)               | B (shift)                |
|-----------------------|------------------------|--------------------------|
| pairs_per             | 20                     | 30                       |
| pair_concentration    | 1.0                    | 0.5                      |
| profile_alpha         | 20.0                   | 8.0                      |
| profile_perturb       | 0.0                    | 0.2                      |
| industry_weights      | (1, 1, 1, 1, 1, 1)     | (2, 1, 1, 2, 1, 1)       |
| corridor_weights      | (1, 1, 1, 1, 1)        | (1, 2, 2, 1, 1)          |
| path_pref_beta        | (2.0, 2.0)             | (1.5, 3.0)               |
| batch_rate            | 0.3                    | 0.45                     |
| batch_size_range      | (2, 49)                | (2, 80)                  |
| batch_tail            | 0.1                    | 0.2                      |
| ultimate_beta         | (3.0, 1.0)             | (2.0, 1.5)               |
| ult_names             | (1, 3)                 | (1, 5)                   |
| amount_logmean        | 10.2                   | 10.6                     |
| amount_pair_sd        | 0.5                    | 0.7                      |
| amount_msg_sd         | 1.2                    | 1.5                      |
| defect_mix            | (0.35, 0.3, 0.2, 0.15) | (0.25, 0.35, 0.25, 0.15) |
| rmt_words_max         | 6                      | 10                       |
| rmt_second_ref        | 0.3                    | 0.5                      |
| name_len              | (4, 17)                | (3, 24)                  |
| route_choice_rate     | 0.3                    | 0.5                      |
| listed_ultimate_share | 0.5                    | 0.6                      |

Each pair draws: creditor industry (weights above) with purpose profile  $\text{Dirichlet}(\alpha p_{\text{industry}})$ , optionally mixed with the uniform distribution; a corridor (weights above); preferred path  $P(\text{pain.001}) \sim \text{Beta}$ ; batching probability rate  $\times 2 \text{Beta}(2, 2)$  and a log-uniform typical file size in the stated range (a share of batched files instead draws 50–200); ultimate presence  $\sim \text{Beta}$  with the stated number of ultimate names; amount log-mean  $\sim N(\mu, \sigma_{\text{pair}})$  with per-message log-s.d.  $\sigma_{\text{msg}}$ ; fixed debtor and creditor names, town, and street. Volume weights are  $\text{Dirichlet}(c \mathbf{1})$  over pairs. Per message: category purpose uniform over five codes; remittance text is an invoice reference, 0 to  $w_{\text{max}}$  vocabulary words, and a second reference with the stated probability; names are random letters of the stated length range. Industry profiles: manufacturing GDDS .45, SUPP .30, TRAD .15, SERV .05, OTHR .05; logistics SERV .40, SUPP .30, TRAD .20, GDDS .05, OTHR .05; retail GDDS

.55, SUPP .25, SERV .10, SALA .05, OTHR .05; services SERV .60, SUPP .15, SALA .15, OTHR .10; NGO CHAR .65, SERV .15, SALA .10, OTHR .10; finance INTC .50, SERV .25, SUPP .15, OTHR .10. Main-run primitive counts (mean per run of 200,000 records, and in the test window): P1a 777 (158); P1b 805 (159); P1c 4560 (906); P1d 950 (195); P2a 2989 (593); P2b 1210 (243); P2c 648 (130); P2d 2781 (541); P3a 199 (41); P3b 401 (82); P3c 896 (180).

## II. SCREENING ABSTRACTION AND CALIBRATION

Purpose weights  $w_p$ : GDDS .10, SUPP .12, TRAD .45, SALA .08, CHAR .38, INTC .30, SERV .18, OTHR .35, absent .32. Corridor weights  $w_c$ : GB-US .05, DE-AE .35, SG-IN .20, US-MX .25, FR-TR .40. Calibration: with  $S = (1 - \pi_0)/(1 - f)$  the name-leg target is  $1 - S^\sigma$  and the score-leg target  $1 - S^{1-\sigma}$ ;  $P_{\text{name}}$  is set so that the mean name-leg probability over an unevaded illicit probe of 4,000 records hits its target (the listed name is in a screened element for only a share of records because the interbank leg does not read ultimate parties), and  $\tau$  is bisected on the score leg. Table II gives the  $3 \times 3$  grid.

## III. FEATURES AND DETECTORS

Message-level vector (11): C1–C4, C5 (purpose novelty against prior pair history), purpose absent, remittance length, address-line count, child count, ultimate absent, any party name of at most five characters. History vector (6 beyond C5): path novelty (share of prior messages on the other path), split deviation ( $|\log n_{\text{ch}} - \text{mean}|/(\text{s.d.} + 0.5)$  over prior files, 0 with fewer than three), ultimate-name novelty, ultimate-presence deviation, amount deviation (same form on  $\log(1 + a)$ ), purpose never seen before (with at least three prior messages). Raw vector (26): one-hot purpose (9 incl. absent), category (6), corridor (6),  $\log(1 + a)$ , child count, ultimate absent, town absent, remittance length.  $D_0 = C1 + \mathbb{I}[C2 > 1]$ ;  $D_1$  as in the paper;  $D_2$  scikit-learn `HistGradientBoostingClassifier`, 150 iterations, other settings default;  $D_3$  `IsolationForest`, 200 trees, on benign training rows;  $D_4 = z(D_1) + z(D_3)$  with training-window statistics;  $D_5$  `LocalOutlierFactor`, 20 neighbours, novelty mode, standardised benign training rows;  $D_6$   $\ell_2$  logistic regression on the standardised raw vector;  $D_7$  sum of the six history deviations standardised on the training window. No class weighting, no tuning, no validation-set selection other than the alert threshold.

TABLE II  
CALIBRATION GRID (SEED 7, 100,000 RECORDS): SOLVED PARAMETERS, TEST AUC, PAIRED SEL RANGE OVER PRIMITIVES.

| $\pi_0$ | $\sigma$ | $P_{\text{name}}$ | $\tau$ | $D_1$ | $D_2$ | $D_5$ | SEL min-max |
|---------|----------|-------------------|--------|-------|-------|-------|-------------|
| 0.3     | 0.4      | 0.126             | 0.828  | 0.780 | 0.919 | 0.785 | 1.15-1.36   |
| 0.3     | 0.6      | 0.185             | 0.894  | 0.780 | 0.866 | 0.784 | 1.14-1.29   |
| 0.3     | 0.8      | 0.240             | 0.996  | 0.759 | 0.826 | 0.735 | 1.17-1.31   |
| 0.45    | 0.4      | 0.236             | 0.708  | 0.784 | 0.906 | 0.805 | 1.25-1.47   |
| 0.45    | 0.6      | 0.337             | 0.782  | 0.774 | 0.877 | 0.771 | 1.26-1.62   |
| 0.45    | 0.8      | 0.429             | 0.896  | 0.765 | 0.840 | 0.741 | 1.36-1.78   |
| 0.6     | 0.4      | 0.365             | 0.608  | 0.785 | 0.921 | 0.806 | 1.43-1.86   |
| 0.6     | 0.6      | 0.508             | 0.690  | 0.767 | 0.884 | 0.773 | 1.38-2.25   |
| 0.6     | 0.8      | 0.629             | 0.814  | 0.778 | 0.863 | 0.754 | 1.59-2.85   |

TABLE III  
ALL EIGHT DETECTORS, MEAN  $\pm$  BETWEEN-SEED S.D. OVER TEN SEEDS; FROZEN-THRESHOLD RECOVERY AND PRECISION AT 1%, REALISED ALERT RATE, EXACT TOP-1% RECOVERY (ANALYSIS METRIC), RECOVERY AT 0.1%; WIDEST PAIR-CLUSTERED 95% HALF-INTERVAL OVER SEEDS FOR AUC AND RECOVERY.

| Det.  | AUC               | AP    | Rec.@1%           | Prec.@1% | Alert rate | Top-1% rec. | Rec.@0.1% | AUC hw | Rec. hw |
|-------|-------------------|-------|-------------------|----------|------------|-------------|-----------|--------|---------|
| $D_0$ | 0.497 $\pm$ 0.002 | 0.044 | 0.009 $\pm$ 0.001 | 0.039    | 0.010      | 0.009       | 0.001     | 0.005  | 0.002   |
| $D_1$ | 0.771 $\pm$ 0.004 | 0.321 | 0.112 $\pm$ 0.005 | 0.499    | 0.010      | 0.112       | 0.022     | 0.013  | 0.015   |
| $D_2$ | 0.886 $\pm$ 0.008 | 0.498 | 0.180 $\pm$ 0.014 | 0.818    | 0.010      | 0.182       | 0.021     | 0.011  | 0.020   |
| $D_3$ | 0.682 $\pm$ 0.009 | 0.079 | 0.022 $\pm$ 0.004 | 0.103    | 0.010      | 0.023       | 0.003     | 0.013  | 0.009   |
| $D_4$ | 0.771 $\pm$ 0.006 | 0.163 | 0.062 $\pm$ 0.007 | 0.276    | 0.010      | 0.062       | 0.018     | 0.012  | 0.012   |
| $D_5$ | 0.775 $\pm$ 0.005 | 0.249 | 0.033 $\pm$ 0.005 | 0.136    | 0.011      | 0.032       | 0.012     | 0.015  | 0.009   |
| $D_6$ | 0.862 $\pm$ 0.009 | 0.436 | 0.155 $\pm$ 0.013 | 0.712    | 0.010      | 0.159       | 0.020     | 0.011  | 0.018   |
| $D_7$ | 0.717 $\pm$ 0.006 | 0.123 | 0.035 $\pm$ 0.004 | 0.205    | 0.008      | 0.046       | 0.003     | 0.015  | 0.009   |

TABLE IV  
PER PRIMITIVE ON THE TEST WINDOW (TEN SEEDS): GAMED TEST RECORDS CARRYING THE PRIMITIVE, PAIRED SEL, ATTEMPT-LEVEL SEL, RESIDUAL SEL AFTER  $D_1$ ,  $D_2$ ,  $D_5$ ,  $D_7$  AT THE FROZEN 1% THRESHOLD, AND THE SHARE ALERTED BY EACH.

|     | $n$ | SEL  | att. | res $D_1$ | res $D_2$ | res $D_5$ | res $D_7$ | hit $D_1$ | hit $D_2$ | hit $D_5$ | hit $D_7$ | hit orig $D_2$ |
|-----|-----|------|------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------------|
| P1a | 158 | 1.48 | 1.43 | 1.16      | 1.15      | 1.49      | 1.45      | 21.7%     | 22.7%     | 0.4%      | 3.2%      | 0.3%           |
| P1b | 159 | 1.38 | 1.38 | 1.18      | 1.27      | 1.39      | 1.37      | 15.1%     | 8.6%      | 0.0%      | 1.4%      | 0.2%           |
| P1c | 906 | 1.41 | 1.40 | 1.33      | 1.24      | 1.39      | 1.38      | 5.9%      | 12.1%     | 2.4%      | 2.9%      | 0.3%           |
| P1d | 195 | 1.47 | 1.42 | 1.13      | 1.30      | 1.47      | 1.44      | 23.3%     | 11.8%     | 1.1%      | 3.3%      | 0.1%           |
| P2a | 593 | 1.41 | 1.43 | 1.39      | 1.27      | 1.33      | 1.36      | 2.6%      | 10.9%     | 6.5%      | 4.5%      | 0.1%           |
| P2b | 243 | 1.32 | 1.32 | 1.28      | 1.17      | 1.28      | 1.31      | 3.9%      | 11.7%     | 3.6%      | 1.3%      | 0.1%           |
| P2c | 130 | 1.27 | 1.26 | 1.19      | 0.85      | 1.18      | 1.26      | 8.2%      | 32.6%     | 8.7%      | 1.9%      | 0.1%           |
| P2d | 541 | 1.44 | 1.47 | 0.96      | 0.75      | 1.43      | 1.39      | 35.3%     | 48.3%     | 2.0%      | 4.2%      | 0.1%           |
| P3a | 41  | 1.56 | 1.53 | 1.54      | 1.43      | 1.50      | 1.56      | 1.5%      | 8.4%      | 4.2%      | 0.9%      | 0.4%           |
| P3b | 82  | 1.44 | 0.94 | 1.41      | 1.39      | 1.44      | 1.37      | 3.2%      | 5.6%      | 0.5%      | 6.1%      | 0.6%           |
| P3c | 180 | 1.59 | 1.60 | 1.58      | 1.43      | 1.56      | 1.44      | 0.8%      | 10.3%     | 2.9%      | 10.5%     | 0.1%           |

#### IV. COMPLETE TEST-WINDOW RESULTS

#### V. PER-PRIMITIVE PAIRED SEL

#### VI. PRE-REGISTERED CONTRADICTION TEST

Labels, outcome, and tests were fixed in `analysis/` before the first full-scale run. Table V reports, for each detector, the variance in per-primitive recovery explained by the contradiction label and by the family, exact permutation  $p$ -values over all relabellings (330 and 11,550), the mean recovery of contradiction-bearing and coherent primitives, Spearman correlations with cost and selection frequency (50,000 permutations), and the leave-one-primitive-out mean absolute error of each grouping against the grand-mean baseline. Per-seed  $R^2$  spread is given as mean  $\pm$  s.d.

#### VII. DEFECT-RATE SWEEP, ABLATIONS, PREVALENCE, SHIFT

#### VIII. LEAVE-ONE-FAMILY-OUT AND ITS MECHANISM

With P1 withheld, an absent ultimate debtor is more common among benign training records (27%) than among the remaining gamed ones: the gamed rate given an absent ultimate debtor is 1.1% against 2.6% when it is present (ten-seed means). The supervised model learns absence as evidence of benignity, so the P1 edits that remove the ultimate debtor score below chance: P1a 0.222 (ultimate absent in 100% of its records), P1c 0.568 (ultimate absent in 20% of its records), P1d 0.210 (ultimate absent in 100% of its records).

#### IX. VALIDITY AND ARTEFACT AUDIT

XSD validation (seed 7): 56,241 of 56,241 instances valid (100.0%), covering every test-window message, every gamed record and every original. By class: P1a 823/823; P1b 790/790; P1c 4619/4619; P1d 965/965; P2a 2937/2937; P2b 1250/1250;

TABLE V  
CONTRADICTION VERSUS FAMILY.

| Det.  | $R_C^2$ | $p$   | mean C | mean not C | $R_{fam}^2$ | $p$   | $\rho_{cost}(p)$ | $\rho_{freq}(p)$ | MAE C | MAE fam | MAE base | $R_C^2$ per seed |
|-------|---------|-------|--------|------------|-------------|-------|------------------|------------------|-------|---------|----------|------------------|
| $D_2$ | 0.04    | 0.567 | 19.8%  | 14.8%      | 0.35        | 0.125 | -0.60 (0.06)     | 0.43 (0.19)      | 0.123 | 0.099   | 0.107    | 0.04±0.02        |
| $D_5$ | 0.00    | 0.955 | 3.0%   | 2.9%       | 0.49        | 0.066 | -0.45 (0.17)     | 0.09 (0.80)      | 0.025 | 0.021   | 0.023    | 0.02±0.02        |
| $D_1$ | 0.05    | 0.524 | 14.2%  | 9.2%       | 0.30        | 0.246 | -0.35 (0.28)     | 0.20 (0.55)      | 0.109 | 0.088   | 0.102    | 0.05±0.02        |
| $D_7$ | 0.05    | 0.542 | 2.8%   | 4.1%       | 0.26        | 0.293 | 0.19 (0.57)      | 0.22 (0.52)      | 0.025 | 0.023   | 0.021    | 0.05±0.05        |

TABLE VI  
TEST AUC AS THE ORDINARY DEFECT RATE  $\eta$  RISES (SEED 7, 100,000 RECORDS).

| $\eta$ | $D_0$ | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ | $D_6$ | $D_7$ |
|--------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.02   | 0.498 | 0.793 | 0.880 | 0.728 | 0.799 | 0.772 | 0.871 | 0.736 |
| 0.05   | 0.499 | 0.808 | 0.896 | 0.736 | 0.818 | 0.799 | 0.881 | 0.746 |
| 0.10   | 0.498 | 0.771 | 0.884 | 0.709 | 0.783 | 0.811 | 0.870 | 0.717 |
| 0.15   | 0.494 | 0.774 | 0.877 | 0.693 | 0.775 | 0.771 | 0.860 | 0.720 |
| 0.25   | 0.490 | 0.742 | 0.870 | 0.658 | 0.736 | 0.778 | 0.851 | 0.707 |
| 0.40   | 0.493 | 0.745 | 0.891 | 0.642 | 0.726 | 0.758 | 0.867 | 0.694 |

TABLE VII  
ABLATIONS (SEED 7, 100,000 RECORDS): TEST AUC / RECOVERY AT THE FROZEN 1% THRESHOLD.

| Factor    | Setting | $D_1$         | $D_2$         | $D_5$         | $D_7$         |
|-----------|---------|---------------|---------------|---------------|---------------|
| family    | P1      | 0.702 / 0.037 | 0.662 / 0.057 | 0.678 / 0.131 | 0.619 / 0.027 |
| family    | P2      | 0.860 / 0.107 | 0.987 / 0.288 | 0.871 / 0.167 | 0.683 / 0.013 |
| family    | P3      | 0.364 / 0.000 | 0.812 / 0.122 | 0.660 / 0.071 | 0.876 / 0.075 |
| budget    | k=1     | 0.659 / 0.076 | 0.778 / 0.144 | 0.655 / 0.055 | 0.670 / 0.015 |
| budget    | k=2     | 0.774 / 0.139 | 0.877 / 0.169 | 0.771 / 0.097 | 0.720 / 0.034 |
| budget    | k=3     | 0.838 / 0.199 | 0.927 / 0.209 | 0.859 / 0.089 | 0.761 / 0.052 |
| knowledge | black   | 0.646 / 0.014 | 0.726 / 0.127 | 0.709 / 0.053 | 0.670 / 0.024 |
| knowledge | grey    | 0.774 / 0.139 | 0.877 / 0.169 | 0.771 / 0.097 | 0.720 / 0.034 |

TABLE VIII  
PREVALENCE SENSITIVITY (SEEDS 7–9, 200,000 RECORDS): GAMED TEST RECORDS, CAPACITY CEILING AT 1%, AND PER DETECTOR AP / PRECISION AT THE FROZEN THRESHOLD / ALERTS PER DETECTED MANIPULATION.

| $\rho$ | gamed (test) | ceiling | $D_1$             | $D_2$              | $D_5$              | $D_7$              |
|--------|--------------|---------|-------------------|--------------------|--------------------|--------------------|
| 0.005  | 180          | 1.000   | 0.147 / 0.114 / 9 | 0.083 / 0.080 / 13 | 0.049 / 0.022 / 49 | 0.022 / 0.035 / 31 |
| 0.010  | 363          | 1.000   | 0.182 / 0.204 / 5 | 0.227 / 0.241 / 4  | 0.097 / 0.045 / 23 | 0.041 / 0.071 / 14 |
| 0.050  | 1813         | 0.221   | 0.321 / 0.495 / 2 | 0.501 / 0.805 / 1  | 0.252 / 0.141 / 7  | 0.127 / 0.213 / 5  |

P2c 675/675; P2d 2762/2762; P3a 209/209; P3b 420/420; P3c 841/841; benign 37970/37970; unevaluated 213/213; original 9029/9029. Invariant violations over 9,029 gamed records: 0. Artefact audit (seed 7, frozen 1% thresholds): benign training rows show 72 distinct remittance lengths, files of up to 200 children, and names of at least 4 characters; 1.2% of gamed test records fall outside those ranges against 0.0% of benign ones, and the share of each detector’s alert mass on such records is  $D_0$  0.0%,  $D_1$  2.2%,  $D_2$  5.4%,  $D_3$  0.0%,  $D_4$  1.0%,  $D_5$  0.0%,  $D_6$  5.4%,  $D_7$  0.3%.

#### X. ETHICS AND RELEASE

None of the eleven operations is disclosed here for the first time; code lists, the harmonised data requirements, and the published screening principles are public, and market-practice guidance already records purpose codes placed in the wrong field. The asymmetry favours the defender, who can map the families onto screened fields, paths, and history features without disclosing payment data. The release is non-operational: all weights are illustrative constants, no value is calibrated against

a real corridor or deployment, and the corpus contains no customer, employer, or production alert data.

#### XI. DOMAIN-RATER QUESTIONNAIRE

For each primitive (definitions only, blind to any result), rate 1–5 with one sentence of reasoning: Q1 actor plausibility; Q2 semantics preservation (same amount, same economic originator and beneficiary); Q3 cost ordering; Q4 schema plausibility; Q5 already described in public guidance; Q6 (yes/no) does the edit make two elements of the message, or an element and the creditor’s profile, inconsistent. Two or three raters with payment-message or financial-crime-control experience; report Cohen’s  $\kappa$  for Q6 and Kendall’s  $W$  for Q1–Q5; Q6 answers replace the pre-specified labels and the contradiction test is rerun unchanged.

TABLE IX  
GENERATOR SHIFT (SEEDS 7–9): FITTED AND FROZEN ON A, SCORED ON B’S TEST WINDOW; B→B AS REFERENCE. AUC / RECOVERY / PRECISION / REALISED ALERT RATE AT THE A-FROZEN 1% THRESHOLD.

| Det.  | AUC A→B | Rec.  | Prec. | Alert rate | AUC B→B | Rec. B→B |
|-------|---------|-------|-------|------------|---------|----------|
| $D_0$ | 0.496   | 0.010 | 0.036 | 0.011      | 0.496   | 0.008    |
| $D_1$ | 0.730   | 0.139 | 0.346 | 0.017      | 0.730   | 0.134    |
| $D_2$ | 0.875   | 0.375 | 0.365 | 0.045      | 0.897   | 0.160    |
| $D_3$ | 0.630   | 0.056 | 0.087 | 0.027      | 0.690   | 0.028    |
| $D_4$ | 0.719   | 0.098 | 0.199 | 0.021      | 0.748   | 0.041    |
| $D_5$ | 0.765   | 0.019 | 0.131 | 0.006      | 0.786   | 0.178    |
| $D_6$ | 0.869   | 0.336 | 0.417 | 0.034      | 0.879   | 0.137    |
| $D_7$ | 0.697   | 0.035 | 0.217 | 0.007      | 0.702   | 0.039    |

TABLE X  
LEAVE-ONE-FAMILY-OUT ON PURE-FAMILY TEST RECORDS ( $k = 1$ , TEN SEEDS), ALL DETECTORS; SUPERVISED MODELS IN → OUT.

| Family | $n$ | $D_0$ | $D_1$ | $D_3$ | $D_4$ | $D_5$ | $D_7$ | $D_2$ / $D_6$ in→out      |
|--------|-----|-------|-------|-------|-------|-------|-------|---------------------------|
| P1     | 773 | 0.497 | 0.551 | 0.615 | 0.598 | 0.514 | 0.576 | 0.633→0.490 / 0.591→0.482 |
| P2     | 734 | 0.502 | 0.825 | 0.589 | 0.769 | 0.834 | 0.715 | 0.976→0.641 / 0.964→0.677 |
| P3     | 130 | 0.504 | 0.383 | 0.450 | 0.419 | 0.538 | 0.851 | 0.550→0.502 / 0.543→0.525 |